

CHRIS BULL

Curriculum Vitae

Ann Arbor, MI — cwbull@umich.edu — <https://cbull81.github.io>

RESEARCH INTERESTS

My interest in epidemiology was borne out of undergraduate academic training in quantitative social science and a subsequent exposure to the challenges of data science approaches to population health during my time in industry. Due to this, I am deeply interested in modeling infectious disease dynamics through a complex systems lens, taking into account not just biological mechanisms but also ecological, climatic, and sociological conditions. In particular, I seek to determine how geospatial heterogeneity may complicate human interaction with the environment and therefore alter disease spread. I hope to explore the possibilities of high-performance computing and machine learning with epidemiologic data to bolster existing analytic frameworks for outbreaks.

ONGOING RESEARCH PROJECTS

Examining the relationship between extreme precipitation and emergency department visits for infectious diseases in the United States, 2006-2021

Led by Dr. Talia Sternbach, under supervision by Dr. Jennifer Head

- Supporting causal pathway discussion and time series model development for multiple infectious disease pathways in all lower 48 U.S. states
- Assuming full responsibility for data management decision-making; building parametrized, scalable SAS scripts for data extraction from a 20% national Medicare claims sample and R scripts for extraction of meteorological raster data at the daily, zipcode level across the lower 48 states
- Assisting with development of distributed lag nonlinear regression models to obtain incidence rate ratios reflecting the association between extreme precipitation and infectious disease outcomes

Investigating the role of temperature volatility and mortality in the United States

Lead author - under supervision by Dr. Jennifer Head

- Reviewing literature to arrive at a robust definition of climate volatility as an epidemiologic exposure
- Summarizing time series mortality data from a 20% national sample of Medicare claims data
- Developing a two-stage distributed nonlinear lag model for assessing the association between temperature volatility and mortality

Developing a cross-village network model of Dengue transmission in Northwestern Ecuador

Lead author - under supervision by Dr. Joe Eisenberg and Dr. Michael Hayashi

- Second-year master's capstone project - developing a metapopulation SEIR model on a cross-village movement network
- Extension of Dr. Eisenberg's long-standing research in Borbon, Ecuador and surrounding communities
- Companion to within-village Dengue transmission model being developed by Dr. Hayashi

EDUCATION

M.S. in Computational Epidemiology and Systems Modeling, May 2027 (exp.)

The University of Michigan, Department of Epidemiology

Graduate Certificate in Complex Systems

Capstone Advisor: Dr. Joe Eisenberg

Cumulative GPA: 3.9

B.S. in Economics, Mathematical Methods in the Social Sciences, June 2022

Northwestern University, Weinberg College of Arts and Sciences

Cumulative GPA: 3.9

ISEN Certificate in Sustainability and Energy

EMPLOYMENT HISTORY

Academic Experience

Research Assistant, Climate-Sensitive Infections Group

August 2026 - Current (full time in the summer, part time during the school year)

University of Michigan, Department of Epidemiology

I currently serve as a research assistant in Dr. Jennifer Head's lab group, funded in part by a research award from the department. My work focuses on examining the effect of extreme weather events on human health outcomes, particularly infectious disease incidence.

Industry Experience

Data Scientist

January 2024 - July 2025

AArete Consulting

I was part of a small data science team that developed Doczy.AI, a first-to-market SaaS tool for processing and extracting metadata from health insurance documents using LLM technology. Utilized by 10+ of the largest healthcare payer firms in America to process millions of documents at scale.

Data Analyst

July 2022 - January 2024

AArete Consulting

I served as a versatile and capable data analyst deployed to various healthcare consulting projects, with responsibilities including building SQL queries for large health insurance claims databases, developing predictive models and clustering analyses for healthcare populations, and conducting ad-hoc data analyses using python.

RELEVANT COURSEWORK

- Epidemiologic Methods
- Data Analysis for Environmental Epidemiology
- Infectious Disease Modeling
- Stochastic Processes
- Probability Theory
- Computer Modeling of Complex Systems
- Machine Learning Methods

TECHNICAL SKILLS

python, R, SQL, SAS, STATA, AWS, LangChain/LLM APIs